Andrea Muolo

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EXPERIENCE

Senior Analytics Engineer and .NET Developer
Systemorph

Dec 2022 - present | Zürich, Switzerland

Developer of the Systemorph Cloud platform and Chief Evangelist.

Organize and conduct presentations to clients, on-site trainings, and workshops.

Tech advocate, lead POCs and implementation of custom web-apps, from data modelling, to business logic and digital workflows.

Junior Analytics Engineer and .NET Developer
Systemorph

Dec 2021 - Nov 2022 | Zürich, Switzerland

Identify and implement data models. Formulate and implement or migrate actuarial methodology. Digital workflows. Everything in C# and MS-SQL.

Postdoctoral Research Fellow
Hebrew University of Jerusalem

Oct 2019 - Nov 2021 | Jerusalem, Israel

Research on: Exact Factorization techniques in molecular physics.

Genuine predictions of the Berry phase. Methods developed in C++20.

Research and Teaching Assistant
ETH Zürich

Dec 2018 - Sept 2019 | Zürich, Switzerland

Research on: Many-body Dirac Equation and relativistic quantum mechanics.

Method development: Pre-Born–Oppenheimer and Tensor Network in C++17.

EDUCATION

Doctor of Sciences
ETH Zürich

Nov 2014 – Dec 2018 | Zürich, Switzerland

Explicitly Correlated Gaussians and the Quantum Few-Body Problem.

Second level Master's Degree in Science and Technologies
Scuola Universitaria Superiore di Pavia (IUSS)

Oct 2009 – Sept 2014 | Pavia, Italy

Focus on transversal and interdisciplinary culture, with a proactive approach.

Face the future with responsibility and confidence to build innovative and sustainable projects and solutions.

BSc and MSc in Physical Chemistry
University of Pavia and University of Cambridge

Sept 2009 – Sept 2014 | Pavia, Italy – Cambridge UK

MSc : A classical dynamic that conserves the quantum Boltzmann distribution.

BSc: Molecular Memories: rotaxane- and catenane-based molecular devices.

MOST PROUD OF

Swiss National Science Foundation (SNSF) Fellowship
Research Grant, October 2019 - Nov 2021

Project Manager - from Jul 2023
Define scope, plan and organize the execution of the POC

PUBLICATIONS SELECTION OF

J. Chem. Theory Comput., 18, 234 (2022)
Quantum Proton Effects from Density Matrix Renormalization Group Calculations

Phys. Rev. A, 102, 022803 (2020)
Analytically projected, rotationally symmetric, explicitly correlated Gaussians with one-axis shifted centers

J. Chem. Phys., 152, 204103 (2020)
Nuclear-electronic all-particle density matrix renormalization group

J. Chem. Phys., 151, 154110 (2019)
H_3^+ as a five-body problem described with explicitly correlated Gaussian basis sets

J. Chem. Phys., 149, 184105 (2018)
Explicitly correlated Gaussian functions with shifted-center and projection techniques in pre-Born-Oppenheimer calculations

Chimia, 70, 244 (2016)
New Approaches for *ab initio* Calculations of Molecules with Strong Electron Correlation

J. Chem. Phys., 142, 134103 (2015)
Boltzmann-conserving classical dynamics in quantum time-correlation functions: "Matsubara dynamics"

COMPUTER SKILLS

- C++20, Armadillo, Boost, Eigen ●●●●●
- OpenMP 4.0, MPI 3.1 ●●●●●
- C# 8, Q#, ML.NET, WPF, Blazor ●●●●●
- Mathematica, LaTex ●●●●●
- Eclipse, Visual Studio, ReSharper, Git ●●●●○
- Linux ●●●●○
- Python, OpenACC ●●●○○

COMPETENCES

Optimization | Parallel computing | SIMD | CI/CD | Quantum computing | Scientific writing | Debugging & Profiling | Teamwork | Fast learning | Data Model | Business App | Problem solving | Particle Physics | Numerical methods | Teaching | Public speaking/presentation | Scrum

LANGUAGES

- Italian (mother tongue) ●●●●●
- English (C1) ●●●●●
- German (A2) ●●○○○

INTERESTS

- Sailing, rowing, swimming, dancing
- Prize novelty, tolerance, and diversity
- Love being with family and friends
- Fueled by curiosity and enthusiasm